

Part A - User and System Investigation (10 minutes)

- Determine the username of the account you are currently using.
whoami
- Display your UID, primary GID, and supplementary groups.
id -u
id -g
id -G
- Display all users currently logged in to the system.
who
- Determine what activities the currently logged-in users are performing.
w
- Examine /etc/passwd and identify the following information for your current account:
Username
UID
GID
Home directory
Login shell
grep "^\$(whoami):" /etc/passwd
- Determine how many user-account entries are present in /etc/passwd.
wc -l /etc/passwd
- Display only the username and UID fields from /etc/passwd.
cut -d: -f1,3 /etc/passwd
- Find all accounts whose UID is 0. Explain the security significance of UID 0.
awk -F: '\$3 == 0 {print \$1, \$3}' /etc/passwd

UID 0 represents the root/superuser account. A user with UID 0 has extremely high privileges and can generally access or modify system resources.

- Compare /etc/passwd and /etc/shadow. Why are password hashes stored in /etc/shadow instead of /etc/passwd?

cat /etc/passwd

/etc/passwd	/etc/shadow
Stores basic information about user accounts.	Stores password hashes and password-related information.
Contains username, UID, GID, home directory, and login shell.	Contains encrypted/hashed passwords and password-aging information.
It is generally readable by all users.	It has restricted permissions and is normally accessible only by root or authorized processes.
The password field usually contains x, indicating that the password hash is stored in /etc/shadow.	Contains the actual password hash.

Password hashes are stored in /etc/shadow because password hashes are sensitive information. The /etc/passwd file needs to be readable by ordinary users and system programs for user and

account information. If password hashes were stored there, any user could access them and potentially perform offline password-cracking attacks.

Part B - Creating and Verifying Users (20 Minutes)

- Create the following users and ensure that a home directory is created for each:

alice, bob, Charlie

sudo useradd -m alice

sudo useradd -m bob

sudo useradd -m Charlie

- Verify that all three accounts have been added to /etc/passwd.

grep -E '^(alice|bob|Charlie):' /etc/passwd

- Determine the UID assigned to each newly created user.

id -u alice

id -u bob

id -u Charlie

- Determine the home directory and default shell assigned to alice.

getent passwd alice

Home directory → /home/alice

Default shell → /bin/bash

- Inspect /home and verify that the required home directories were created.

ls -ld /home/alice /home/bob /home/Charlie

- Set passwords for all three accounts.

sudo passwd alice

sudo passwd bob

sudo passwd Charlie

- Switch from your current account to alice using a login shell. Verify the change using an appropriate command, and then return to your original account.

su - alice

whoami

pwd

exit

Part C - Group Management and Access Control (30 Minutes)

- Create two groups:

security and developers

sudo groupadd security

sudo groupadd developers

- Add alice and charlie to the security group.

sudo usermod -aG security alice

sudo usermod -aG security Charlie

- Add bob to the developers group.
sudo usermod -aG developers bob
- Without switching accounts, verify the group membership of all three users.
groups alice
groups bob
groups Charlie
- Display the entries corresponding to security and developers from /etc/group.
grep -E '^(security|developers):' /etc/group
- Add bob to the security group without removing him from the developers group.
sudo usermod -aG security bob
- Verify that bob now belongs to both groups.
groups bob